Session 11B Teamwork

Close the same gap on Workspaces, then verify it in MongoDB

Outcome
Workspaces now uses persisted Mongoose documents, validates integer route IDs, supports validated name/slug filters, and seeds the initial records only when the collection is empty.

What changed

1. Defined the Workspace MongoDB document

Added **src/workspaces/schemas/workspace.schema.ts** with the Project schema shape: required numeric id, required name, required slug, and description defaulting to an empty string. The slug takes the role of Project.key. SchemaFactory creates the Mongoose schema.

2. Registered the model and connected dependency injection

Added MongooseModule.forFeature([{ name: Workspace.name, schema: WorkspaceSchema }]) to WorkspacesModule. WorkspacesService injects the matching model token with @InjectModel(Workspace.name). This registration is what makes the injected model provider available at runtime.

3. Replaced the in-memory array with persisted operations

WorkspacesService now implements OnModuleInit. At startup it counts documents and inserts the two starter workspaces only when the collection is empty. Reads, creation, updates, uniqueness checks, and slug lookup now query or save Mongoose documents; generated numeric IDs are one greater than the current largest ID.

4. Added validated workspace query parameters

Added FindWorkspacesQueryDto with optional string name and slug fields, following the Projects query DTO. WorkspacesController passes @Query() to the service. The service applies a case-insensitive regular-expression name filter and an exact slug filter; with no filters it returns every persisted document.

5. Closed both integer route-parameter gaps

Both findOne and update now use ParseIntPipe and type their IDs as numbers. Invalid values such as abc are rejected as HTTP 400 before service/database work; valid numeric IDs retain their existing behavior.

6. Preserved rejection of unknown query keys

The existing global ValidationPipe in src/main.ts uses whitelist and forbidNonWhitelisted. Because the new query DTO only declares name and slug, a query such as ?foo=bar returns HTTP 400 rather than being silently ignored.

Why controller and service changes must be paired

Change	What must match
@InjectModel in the service	The service requests a model provider by token. WorkspacesModule must register the same Workspace model with MongooseModule.forFeature; without it Nest cannot resolve the dependency at application startup.
@Query() in the controller	The controller now calls findAll(query). The service must accept that DTO argument and implement its filters; leaving the old no-argument method makes the controller/service contract inconsistent and would ignore the requested filtering.

Verification results

Acceptance check	Observed result
GET /workspaces/abc	400 Bad Request - ParseIntPipe
PATCH /workspaces/abc	400 Bad Request - ParseIntPipe
GET /workspaces/1	200 OK - Acme Robotics document
GET /workspaces	200 OK - documents loaded from MongoDB
GET /workspaces?name=nova	200 OK - Nova Labs only
GET /workspaces?slug=acme-robotics	200 OK - Acme Robotics only
GET /workspaces?foo=bar	400 Bad Request - unknown query rejected
POST /workspaces (Growth)	201 Created - id 3, slug growth
mongosh query on b4f-board.workspaces	Growth document found with the same MongoDB _id

Independent persistence proof

The API was started with the local b4f-board database for this verification. The workspace was created over HTTP, then read directly from the workspaces collection in a separate mongosh command. Both outputs carry the same MongoDB _id.

```
curl -i -X POST http://localhost:3000/workspaces \
  -H "Content-Type: application/json" \
  -d '{"name":"Growth","slug":"growth"}'
```

curl response - HTTP 201 Created


```
{"id":3,"name":"Growth","slug":"growth","description":
"","_id":"6accdd95f53dee5db5e023ab","__v":0}
```


mongosh direct collection query

```
db.workspaces.find({slug:/^growth$/})
```


```
{
  _id: ObjectId('6accdd95f53dee5db5e023ab'),
  id: 3,
  name: 'Growth',
  slug: 'growth',
  description: '',
  __v: 0
}
```


Repeatable database check

```
mongosh --quiet 'mongodb://localhost:27017/b4f-board' --eval 'db.workspaces.find({
slug:/^growth$/}).forEach(printjson)'
```

Build and environment notes

The API build and TypeScript check passed, and ESLint passed on all changed Workspace files. The installed TypeScript compiler is 5.9.3, so the stale ignoreDeprecations value was adjusted from 6.0 to 5.0. The build config disables incremental emission for the production build: Nest removes dist before compiling, and stale incremental metadata otherwise caused a successful build to leave dist/main.js missing.

The existing api/.env was not modified. Its database path differs from the exercise database, so the API process for this proof used MONGODB_URI=mongodb://localhost:27017/b4f-board. Start it the same way for repeatable checks, or change your local environment deliberately if b4f-board should be the permanent target.

Changed implementation files: workspaces.controller.ts, workspaces.service.ts, workspaces.module.ts, dto/find-workspaces-query.dto.ts, schemas/workspace.schema.ts. Build configuration: tsconfig.json and tsconfig.build.json.